

Joshua Oh Yu Qiao

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Available: 12-week placement, 25 January – 23 April 2027

SUMMARY

Third-year Electronic and Electrical Engineering student (CGPA 3.91) with hands-on experience designing, testing and debugging hardware — three projects carried from circuit to working prototype in embedded firmware, machine vision and bench measurement. Seeking an internship in test, product, equipment or firmware engineering.

EDUCATION

Bachelor of Electronic and Electrical Engineering with Honours Apr 2024 – Apr 2028

Sunway University, Subang Jaya

- Current CGPA: 3.91 / 4.00 — Dean's List: Sep 2024, Apr 2025, Sep 2025 semesters
- Jeffrey Cheah Entrance Scholarship recipient

Cambridge A-Levels — 3A (Mathematics, Physics, Chemistry) Jul 2022 – Nov 2023

INTI International College Subang — Top Student, A-Level (2023) & AS-Level (2023)

Sijil Pelajaran Malaysia (SPM), Science Stream — 8A, 1B Jan 2017 – Mar 2022

SMK Bandar Sri Damansara 1, Kuala Lumpur

ENGINEERING PROJECTS

AI Waste-Sorting Prototype

Sustainable Engineering Design, group project

- Trained and deployed a custom **YOLOv11s** model on a self-collected 110-image dataset to classify metal, paper and plastic in real time, achieving **0.98 mAP@0.5** and **0.96 F1** at 0.63 confidence.
- Integrated the vision pipeline with an ESP32 over HTTP to drive a conveyor and dual servo diverters, with three **HX711 load cells** for per-bin weight logging, a Blynk dashboard and automated bin-full alerts.
- Modelled the mechanical assembly in SOLIDWORKS and 3D-printed it in PLA (242 g, 23 × 18 × 19 cm); designed the circuit in EasyEDA and assembled it on perfboard.

Solar-Powered IoT Irrigation Controller

Project Management & Engineering Design — Project Manager and Software Engineer, 5-person team

- Characterised the Li-ion power system at **0.89 W idle / 12.8 W actuating** (~1.7 days runtime per charge); diagnosed and suppressed **solenoid back-EMF** with a flyback diode after it caused relay chatter.
- Built the firmware on an ESP8266 in C++ (VS Code / PlatformIO / Git), reading a DS18B20 probe to drive relay-switched solenoid valve actuation above a 38 °C threshold — preventing thermal shock to crops — with a Blynk dashboard for remote auto/manual control and NTP-synced logging.
- Led the team to delivery on a 13-week CPM/Gantt schedule and **9% under budget**, running critical-path analysis and budget variance reporting.

PIPER — Privacy-Preserving Dementia Assistive Robot

Integrated Design Project, 4-person team — System Architect & Firmware Lead · prototype build underway

- Architected a two-tier system pairing an ESP32 with a Python/Flask backend running local Whisper STT, Gemini LLM and Piper TTS over an SQLite event-memory database, enabling the robot to recall the user's daily events (medication, movement, appliance state).
- Ran a comparative literature review of fall-detection sensing methods and selected a **24 GHz mmWave radar (LD2450)** over camera-based alternatives, eliminating video capture in the home while preserving day/night reliability.
- Built Telegram caregiver alerting with a 10-second cancellation window and MQTT smart-home control across two ESP32 nodes, designed to a **<3 s voice-response** and **<10 s alert budget**.

TECHNICAL SKILLS

- **Programming:** C/C++, Python (NumPy, Pandas, OpenCV, PyTorch/Ultralytics), MATLAB & Simulink, Git/GitHub, VS Code, PlatformIO, Arduino IDE
- **Embedded & Hardware:** ESP32 / ESP8266 (NodeMCU), sensor interfacing (mmWave radar, DS18B20, HX711 load cells, I²S MEMS microphones), servo & relay actuation, I²C / I²S / UART / OneWire, voltage regulation and power budgeting, soldering and perfboard prototyping, oscilloscopes and multimeters
- **IoT, Cloud & AI:** MQTT, Flask REST APIs, SQLite, HTTP/REST device communication, LLM and speech-pipeline integration (Gemini, Whisper, Piper TTS), Blynk, ThingSpeak, SMTP/Telegram alerting, real-time dashboards and data logging
- **Design & Simulation:** SOLIDWORKS, OnShape, EasyEDA, NI Multisim, FDM 3D printing (PLA/ABS)
- **Engineering Practice:** Project management (CPM, Gantt, budget variance analysis), decision-matrix design selection, laboratory instrumentation, technical report writing
- **Languages:** Bahasa Malaysia, English, Mandarin, Cantonese (fluent); German (basic)

WORK EXPERIENCE

Assistant Head, Events & Social Media — Sales Support Team (Part-time) Feb 2023 – Apr 2024

INTI International College Subang, Subang Jaya

- Led a team of 8 to 12 student helpers in various events (Mid Valley Higher Education Fair, INTI Open Day, WOW Event).
- Involved in the recruitment process for events with the HR Team.
- Conducted meetings and training sessions for student helpers ahead of events.
- Conducted Data Entry and rescue calls to help with leads collection.